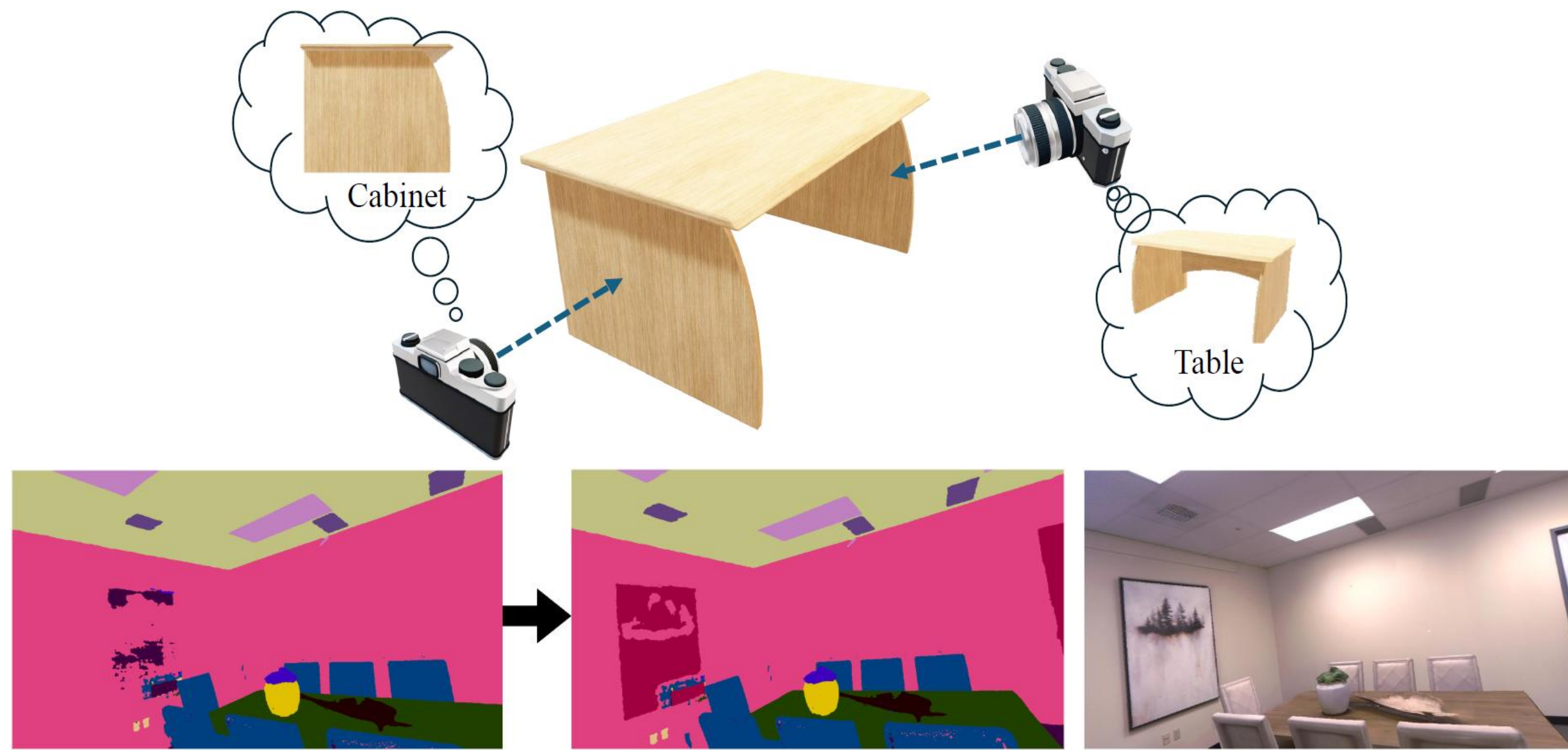


Introduction

Motivation

- Semantic predictions are view-dependent, causing the same 3D region to receive inconsistent predictions under occlusion or partial observation.
- Existing uncertainty-guided active mapping mainly evaluates individual views and does not explicitly exploit **cross-view semantic inconsistency** for exploration.

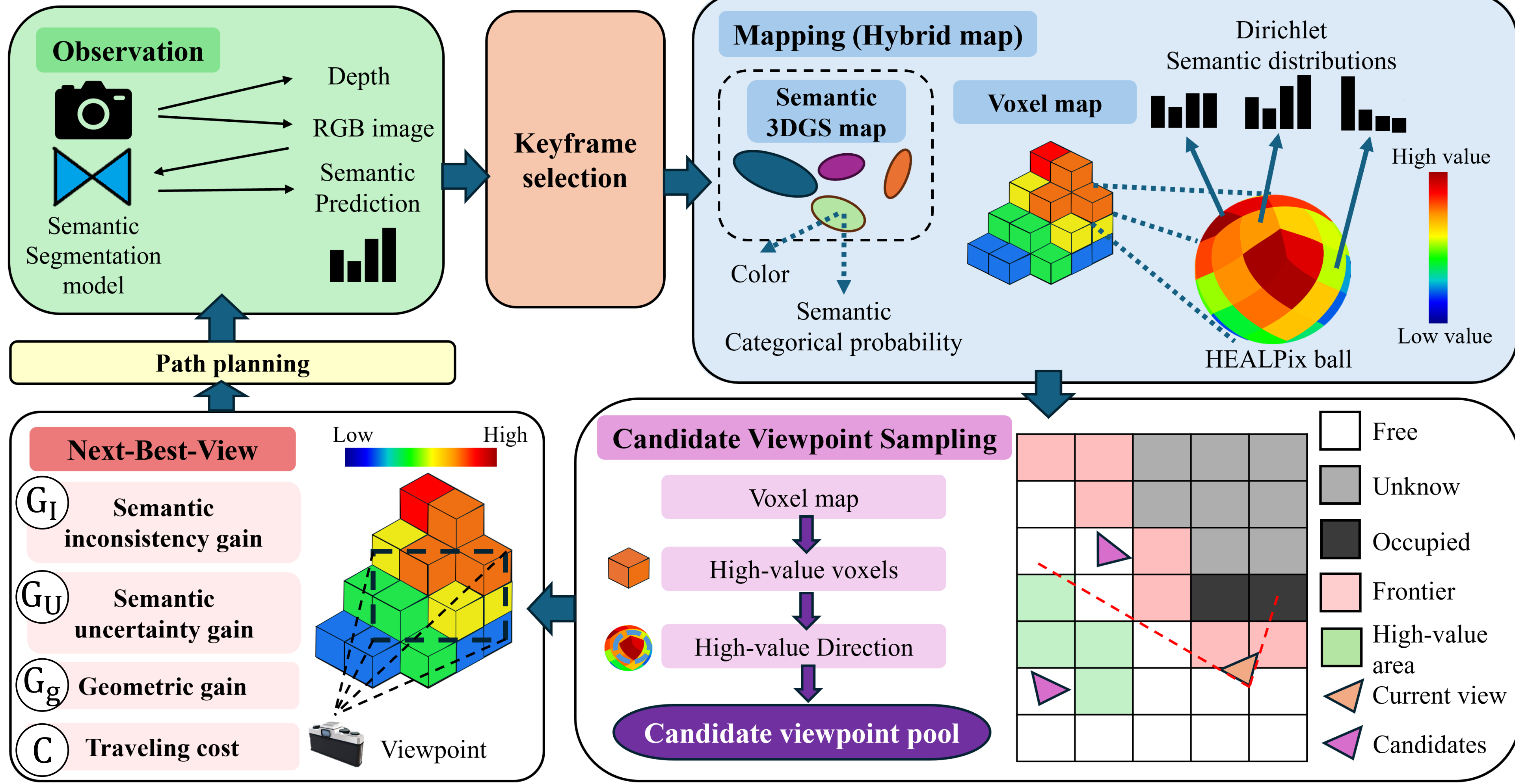


Contributions

- Preserve direction-conditioned semantic evidence across viewpoints.
- Convert cross-view inconsistency into exploration heat.
- Select viewpoints based on semantic inconsistency to reduce semantic ambiguity

Method

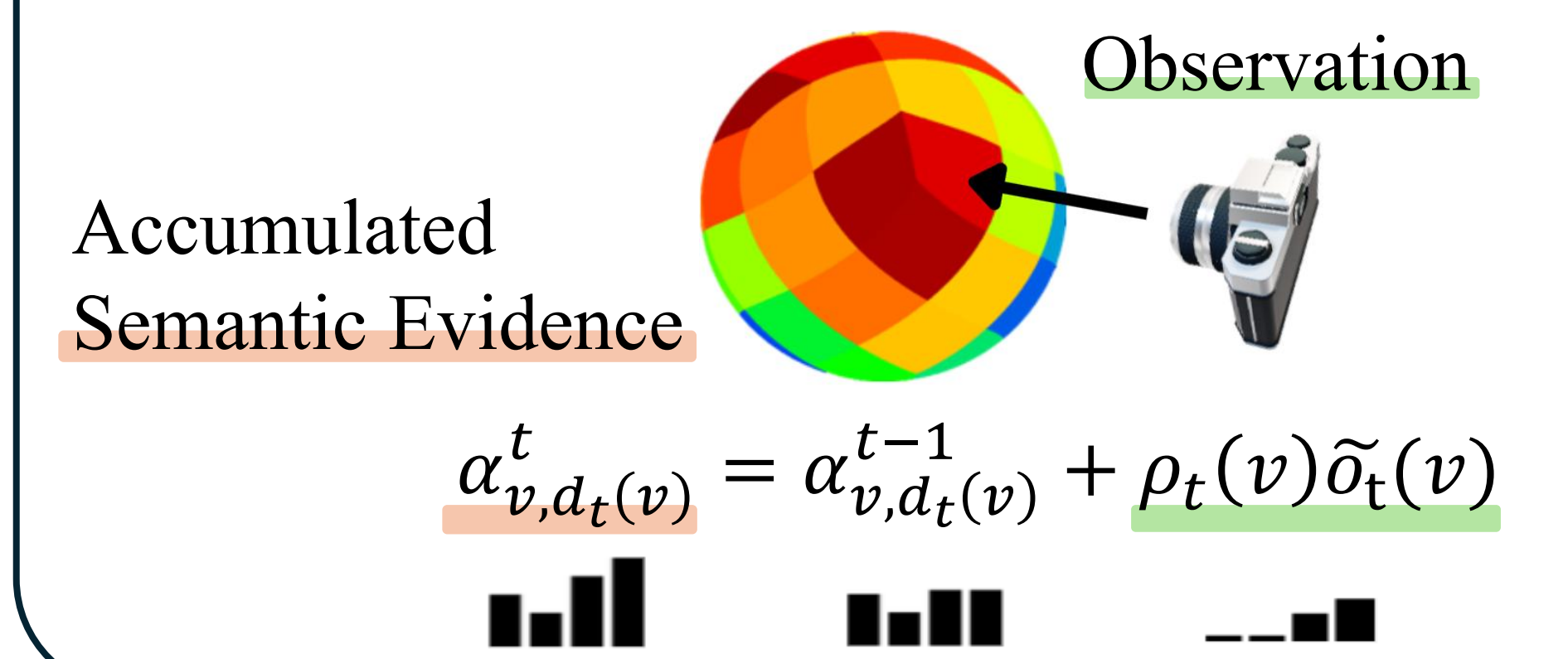
1. Pipeline



2. Hybrid map

Map	Role
Gaussian	Dense reconstruction and rendering
Voxel	Occupancy, directional semantic evidence, and exploration statistics

Directional Semantic Evidence



3. Directional Exploration Heat: $H_{v,d}$

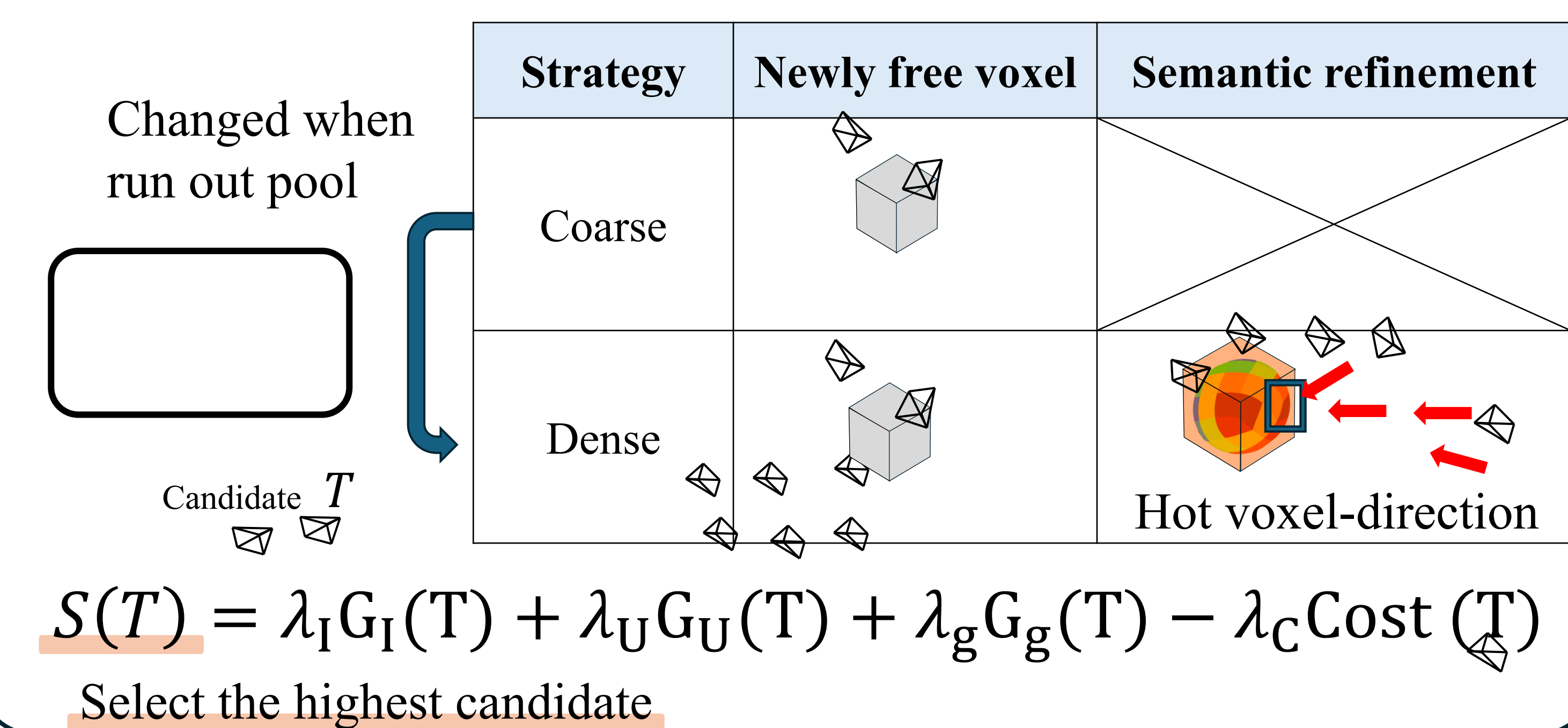
$$D_{v,d} = \sum_{d' \in \mathcal{N}(d)} \frac{JSD(\mu_{v,d}, \mu_{v,d'})}{|\mathcal{N}(d)|}$$

$$H_{v,d} = m_{v,d} D_{v,d} \frac{1}{S_{v,d} + 1}$$

v : voxel, d : direction bin in v

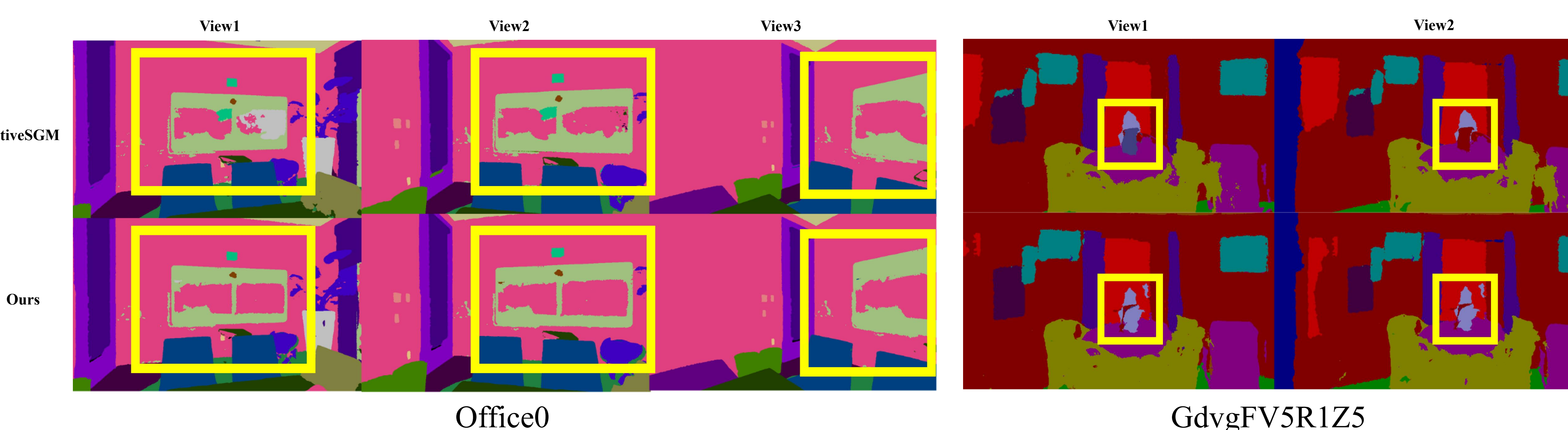
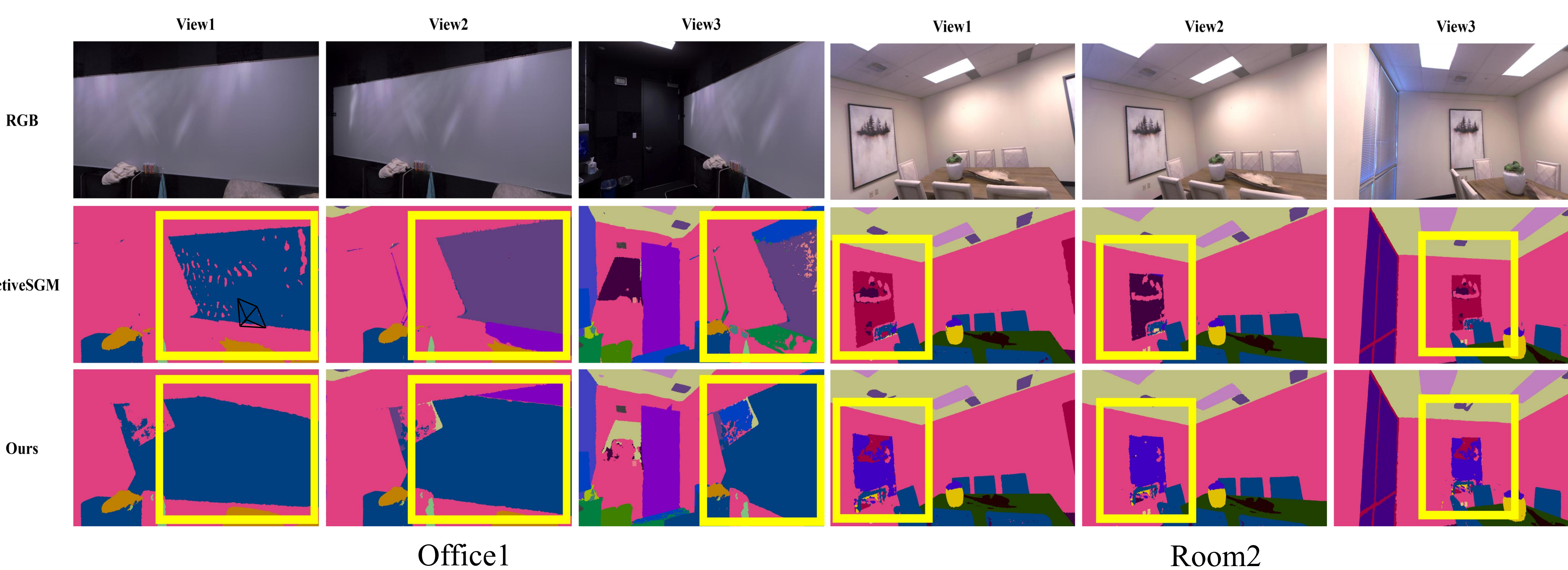
Term	Meaning
$D_{v,d}$	Directional semantic inconsistency
JSD	JS-Divergence
$\mathcal{N}(d)$	Valid adjacent directions of d
$m_{v,d}$	Feasible observation direction mask
$S_{v,d}$	Accumulated evidence strength

4. Candidate Viewpoint Sampling



Results

Novel-view comparisons complement per-view mIoU/mAP by evaluating semantic stability across viewpoints. ActiveSIGMA maintains more consistent semantic predictions than ActiveSGM.



ActiveSIGMA improves semantic mapping on both Replica and MP3D while maintaining comparable geometric and photometric reconstruction quality.

Dataset	Method	Avg. Steps	PSNR $\uparrow$	RMSE $\downarrow$	mIoU $\uparrow$	mAP $\uparrow$
Replica	SGS-SLAM	2000	27.13	0.0708	80.45	89.98
Replica	ActiveSGM	783	30.46	0.0138	84.48	93.22
Replica	ActiveSIGMA	768	30.83	0.0124	84.86	94.33
MP3D	ActiveSGM	2325	26.93	0.0190	74.54	86.72
MP3D	ActiveSIGMA	2007	26.80	0.0175	74.69	86.88

Ablation Study

Variant	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$	mIoU $\uparrow$	mAP $\uparrow$
w/o geom. gain	30.26	0.9522	0.1553	0.0164	84.35	93.05
w/o sem. incons. gain	30.61	0.9565	0.1520	0.0120	84.10	94.02
w/o sem. uncert. gain	30.70	0.9572	0.1455	0.0139	84.58	94.24
w/o semantic gains	30.01	0.9525	0.1530	0.0114	83.94	93.14
ActiveSIGMA	30.83	0.9579	0.1465	0.0124	84.86	94.33

Conclusion

- Cross-view semantic inconsistency is an informative exploration signal.
- Direction-conditioned evidence guides where and from which direction to acquire additional observations.
- ActiveSIGMA improves semantic mapping on Replica and MP3D and producing more consistent novel-view semantic predictions.